Lab - Running a Simple Container (Docker)
To Begin with the Lab

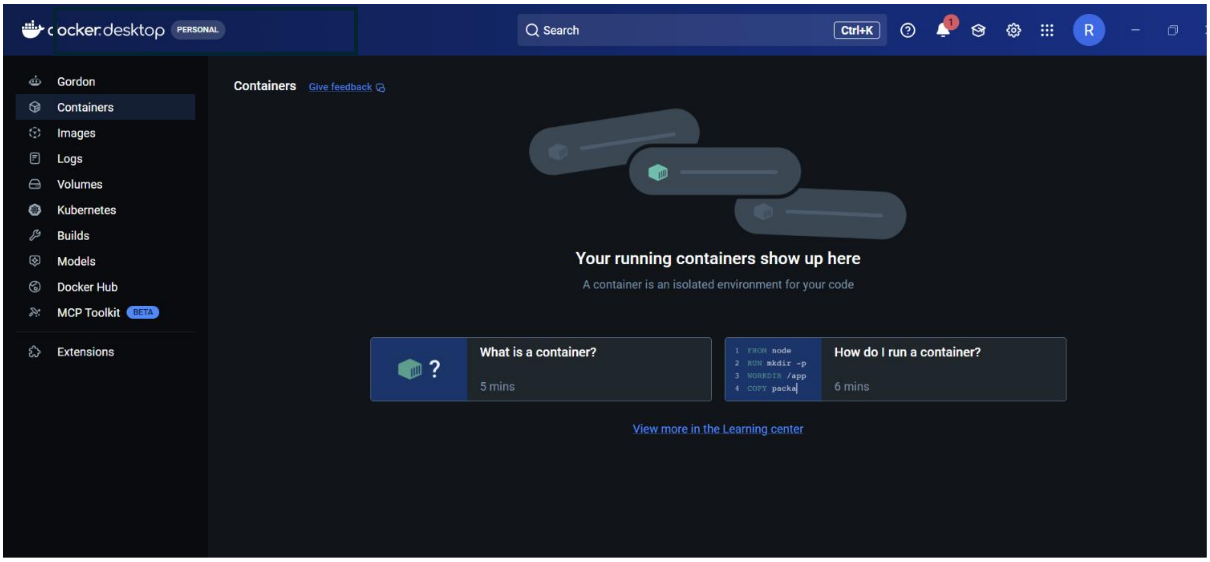
Summary
This lab demonstrates how to run a Docker container from an existing Docker image. It explains the role of Docker Desktop, Docker Hub, Docker images, containers, and how to use Docker commands to download and run an Nginx web server inside a container.

Understanding
Docker Desktop provides a graphical interface (GUI) and installs the Docker Engine (Docker runtime) on your computer. The Docker Engine is responsible for downloading Docker images, creating containers, and managing them.
A **Docker Image** is a read-only template that contains everything needed to run an application. A **Docker Container** is a running instance of that image.
In this lab, the **Nginx** image is downloaded from **Docker Hub**, and Docker creates a container from that image. The container runs the Nginx web server without installing Nginx directly on the operating system.
Port mapping allows users to access applications running inside the container from the host machine. For example, mapping **Host Port 8080** to **Container Port 80** makes the Nginx website available at http://localhost:8080.
Example

- Docker Image: nginx:latest
- Docker Container: Running instance of the Nginx image
- Host Port: 8080
- Container Port: 80
- Access URL: http://localhost:8080

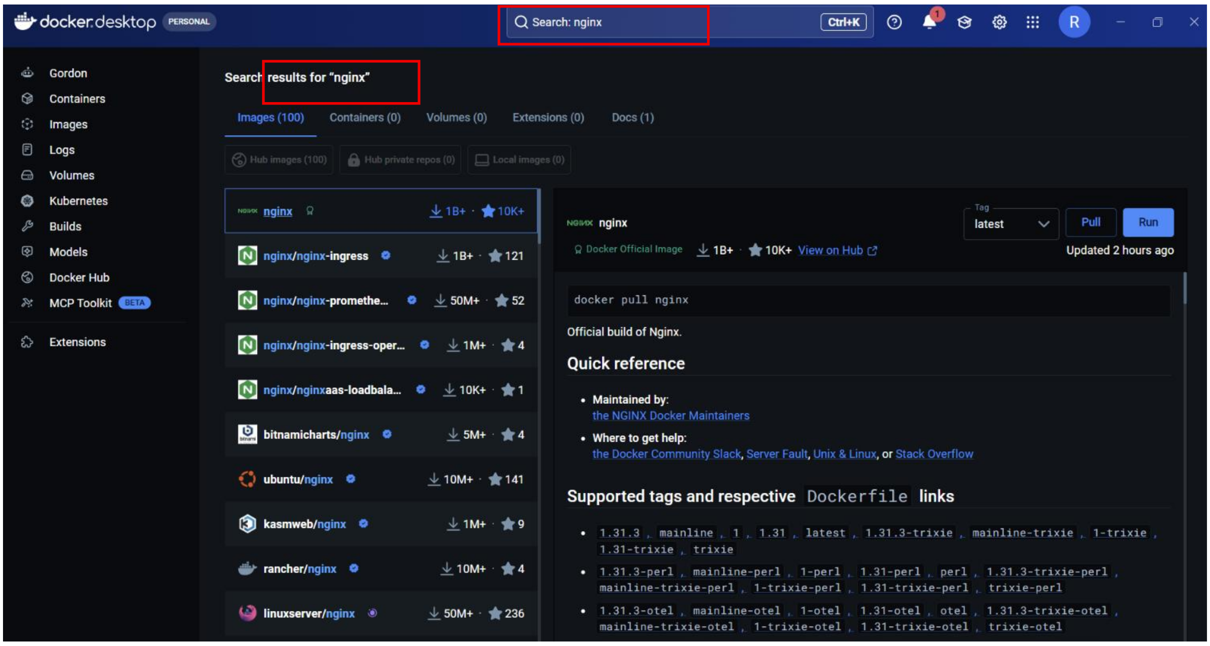
Lab Steps
Step 1: Open Docker Desktop
- Launch Docker Desktop.
- Wait until Docker Engine is running.
- Ensure Docker status shows **Running**.

Step 2: Create a Docker Hub Account
- Open **https://hub.docker.com**.
- Sign up using Email or Google account.
- Verify your email.
- Sign in to Docker Hub.



Step 3: Search for the Nginx Image

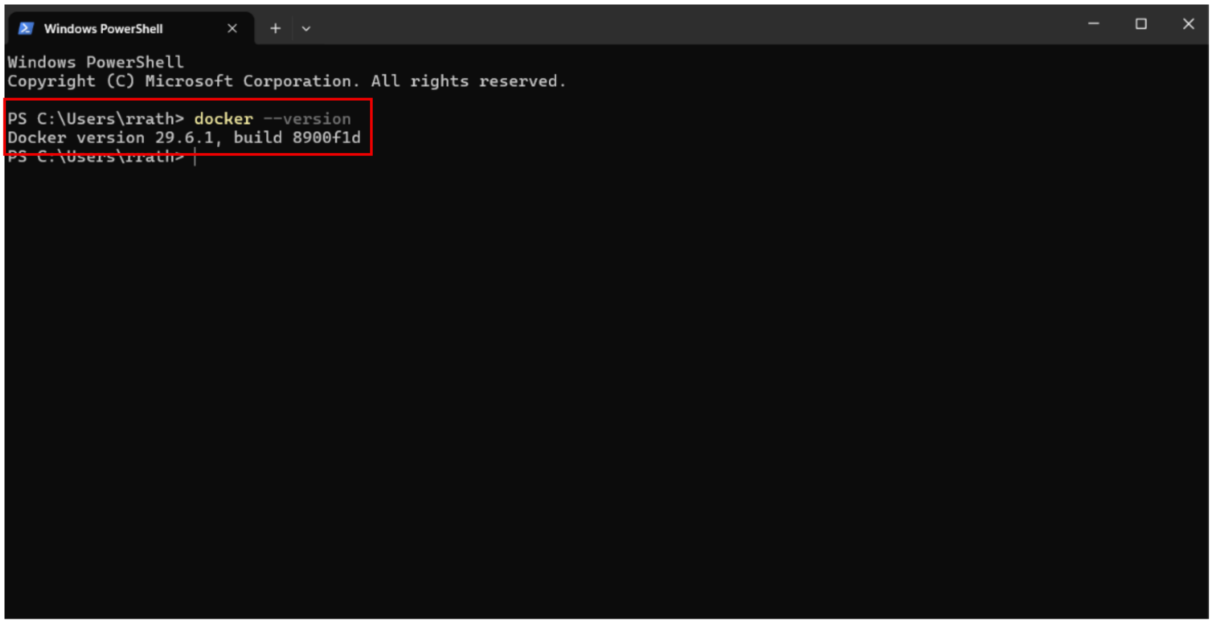
- Search for **nginx**.
- Open the **Official Nginx Image**.
- Review available image tags.
- Note the pull command.



Step 4: Open Command Prompt

- Open Command Prompt or PowerShell.
- Verify Docker is installed.

```
docker --version
```



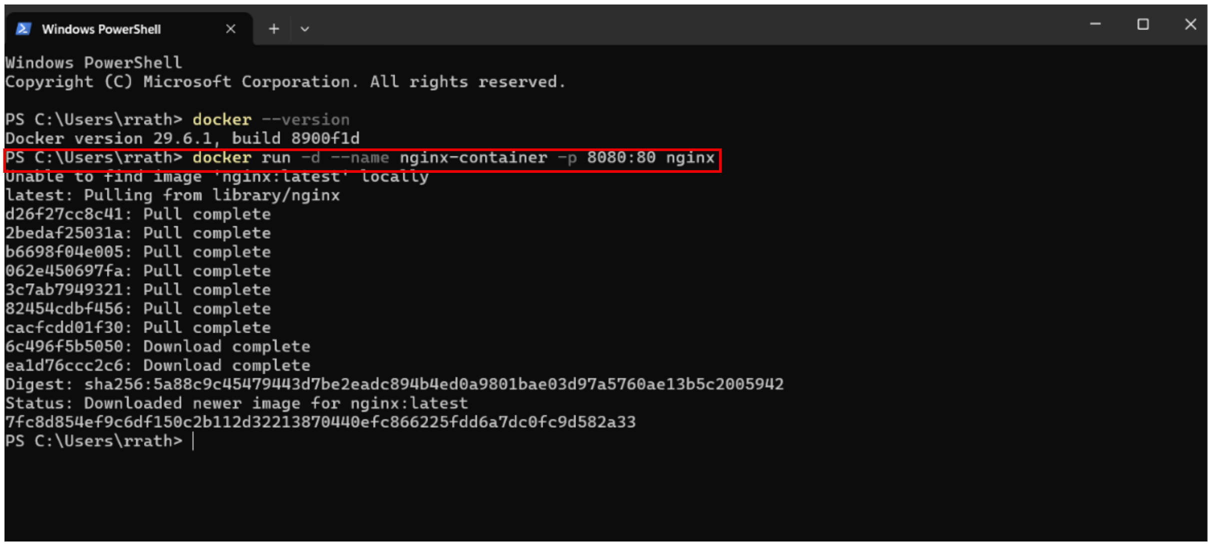
Step 5: Run an Nginx Container

Execute the following command:

```
docker run -d --name nginx-container -p 8080:80 nginx
```

Command Explanation

- docker run → Creates and starts a container.
- -d → Runs the container in background (Detached mode).
- --name nginx-container → Assigns a container name.
- -p 8080:80 → Maps Host Port 8080 to Container Port 80.
- nginx → Docker image name.



Step 6: Verify Running Containers

List running containers.

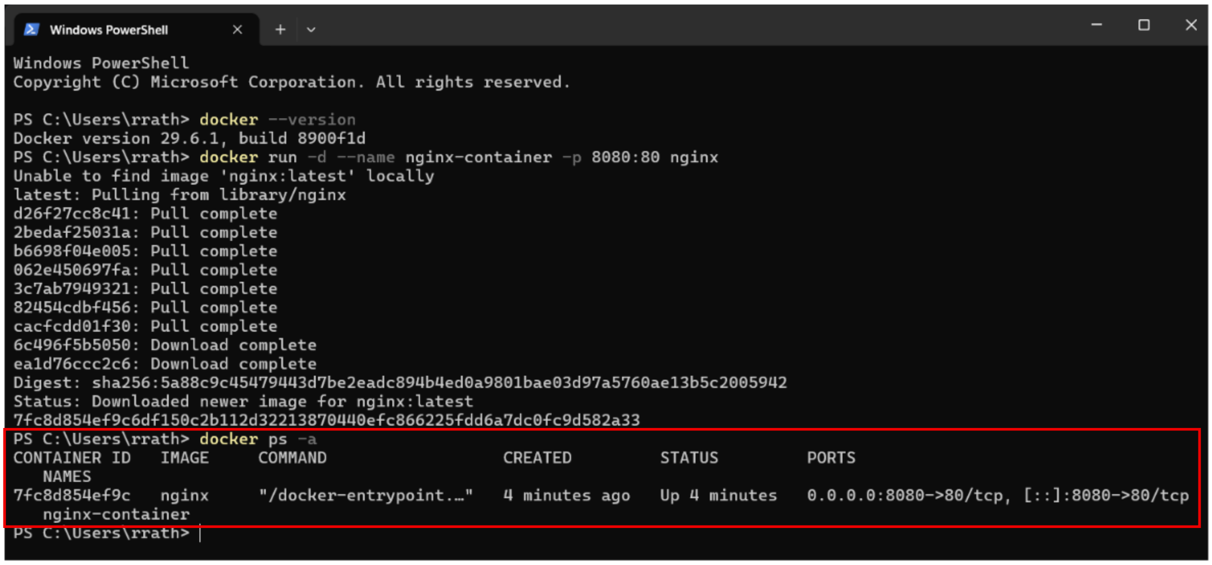
Execute the following command:

```
docker ps -a
```

Expected output includes:

- Container ID
- Image Name

- Container Name
- Status
- Port Mapping



Step 7: View Downloaded Images

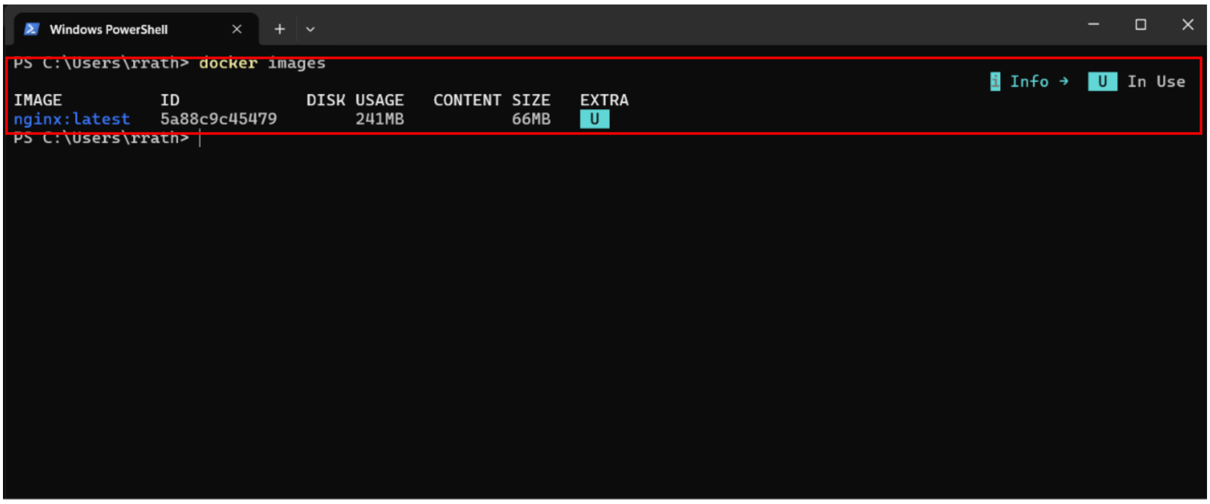
Display all images stored locally.

Execute the following command:

docker images

This shows:

- Repository
- Tag
- Image ID
- Size



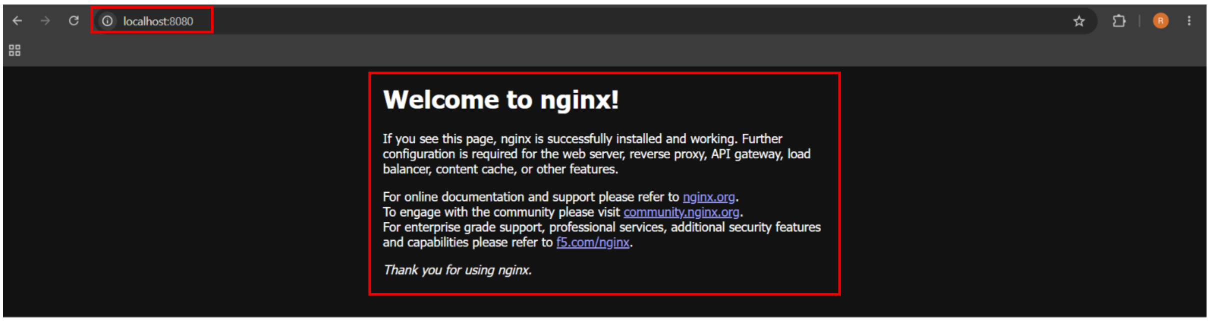
Step 8: Access the Nginx Website

Open a **web browser**.

Visit:

http://localhost:8080

The default **Welcome to nginx!** page should appear.

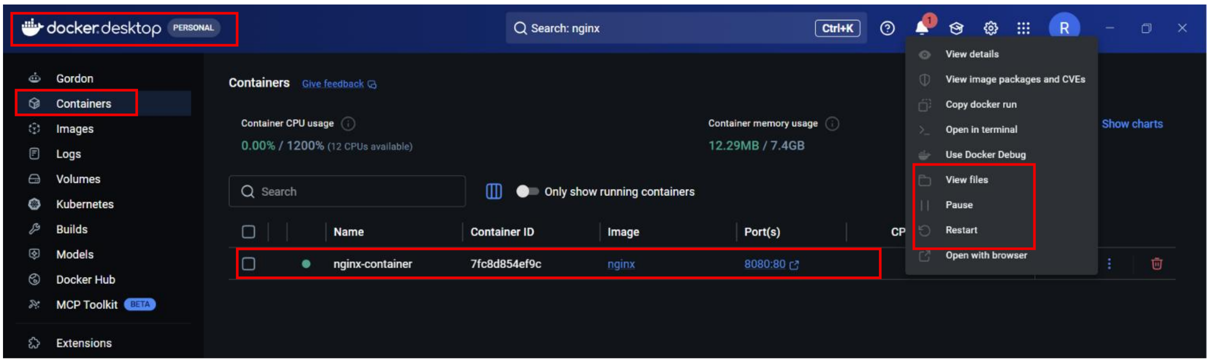


Step 9: Manage Containers Using Docker Desktop

Open Docker Desktop.

Under **Containers**, you can:

- View running containers.
- Stop a container.
- Restart a container.
- Pause a container.
- Delete a container.



Step 10: View Docker Images

Go to the **Images** section in Docker Desktop.

Verify that the **nginx** image is available locally.